

The Clay Navier–Stokes Problem:

Formulation and Solution

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Abstract

The official formulation of the Clay Millennium Problem on the Navier–Stokes equations asks whether smooth initial data give a solution that stays smooth for all time, or whether the velocity can blow up in finite time. The alternative is exhaustive only if those are the two possibilities. They are not. The solutions of physical interest are *turbulent*: not smooth, because velocity gradients grow as $\nu^{-1/2}$ and are enormous at the Reynolds numbers of any real flow, and not singular, because the velocity itself stays bounded. A dichotomy that omits the only case that matters cannot be answered in a way that says anything about fluids, and the case it does cover — persistent smoothness — is either trivial or false depending on how it is read.

This article argues that the problem as posed is mis-posed, and states what should replace it. The replacement is Hadamard’s criterion: not global smoothness, but *wellposedness*. A solution concept earns physical meaning when the quantities computed from it are stable under perturbation. For turbulent solutions those quantities are not pointwise velocities, which are indeterminate, but mean-value outputs such as drag and lift, which are not. We set out the distinction, the evidence that turbulent solutions are wellposed in outputs while smooth solutions are not, and the computational record behind it: a Galerkin method whose only turbulence model is its own residual, error control by duality with computed stability factors, and mean drag and lift converging to a few per cent on meshes on which the velocity field does not converge and never will. The reformulated problem has a solution, and the solution is that turbulence is what Navier–Stokes has to say.

1 The official formulation

The Clay Mathematics Institute problem, as set out by Fefferman [1], concerns the incompressible Navier–Stokes equations

$$\dot{u} + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0, \quad (1)$$

on $\mathbb{R}^3$ or a periodic box, with $u(\cdot, 0) = u^0$ smooth and divergence free, and with the viscosity ν normalised to one. Four statements are offered, of which a proof of any one settles the prize: existence and smoothness on $\mathbb{R}^3$; the same with periodic data; or a counterexample to either — a solution which is initially smooth and whose velocity becomes unbounded in finite time.

The structure of the question is therefore a dichotomy. Either the solution is smooth for all time, or it is not, and the only stated way for it not to be is blowup. *Tertium non datur*.

2 Why the dichotomy is empty

The physical solution is neither. A fluid at a Reynolds number of interest — 10^6 and upward for an aircraft, a ship, a weather system, a pipe — is turbulent. Turbulent flow is not smooth in any useful sense: the velocity varies on a length scale that shrinks with viscosity, and the gradients scale as

$$|\nabla u| \sim \nu^{-1/2}, \quad (2)$$

so that at $\nu = 10^{-6}$ they are of order 10^3 times the velocity scale, and at the viscosity of air over a wing, larger still. Nor is it singular: the velocity stays bounded, the kinetic energy stays finite, and nothing goes to infinity anywhere. Kolmogorov’s scaling, in the form Onsager gave it, puts the velocity field at Hölder continuity $1/3$ — continuous, with unbounded derivatives in the limit. That is precisely the case the official alternative does not contain.

The smooth case is trivial or false. If smoothness is to be read as “the solution exists and is smooth for a short time”, the statement is classical and was not worth a prize. If it is to be read as “the solution stays smooth for all time”, then for the flows of interest it is false, and not because of blowup: a smooth solution of (1) at small viscosity is *unstable*. Perturb potential flow around a bluff body and it does not stay potential flow; it develops rotational separation and becomes turbulent. The smooth solution exists as a formal object and is not observed, because nothing keeps a flow in it. So the first horn of the dichotomy describes solutions that are mathematically available and physically absent, and the second describes solutions that do not occur.

The normalisation is legitimate, and that makes the objection worse. It might be thought that setting $\nu = 1$ removes the phenomenon, since all the structure in (2) is structure in ν . It does not. Navier–Stokes rescales as

$$v(x, t) = \alpha u(\beta x, \alpha \beta t), \quad \nu' = \nu \beta / \alpha,$$

so any viscosity can be set to one by rescaling the data, and small ν with moderate data is the same problem as $\nu = 1$ with large data. Fefferman asks about *all* Schwartz-class initial data, so arbitrarily high Reynolds number is inside the hypotheses.

The consequence is not that the formulation is saved but that it is worse off. Turbulent data are covered, and a turbulent solution that never blows up is, in the technical sense, *smooth*. So the first horn of the dichotomy would come out true — and say nothing whatever, because smoothness does not distinguish laminar flow from turbulence. A theorem asserting global smoothness with derivatives bounded by an iterated exponential is a theorem about a fluid in the way that a bound of $10^{10^{10}}$ metres is a statement about the diameter of this room. The vacuity is in what the dichotomy asks for, not in the normalisation.

The quantitative results confirm rather than relieve this. Work in the analytic tradition has produced conditional regularity bounds of the form: if the velocity is bounded by A , then the derivatives are bounded by an expression exponential in A iterated several times. As a theorem this is real. As a statement about a fluid it is empty: a bound of the size $\exp \exp \exp A$ is not a bound that any flow can be said to respect, and no measurement could distinguish it from no bound at all. Tartar’s objection to the problem’s framing — that the selected questions have almost no physical content, and appear to have been chosen by specialists in harmonic analysis rather than by anyone studying continuum mechanics — is, on this evidence, well taken [3].

Non-smooth and non-singular, quantitatively. The two halves of the third case are one scaling seen from either end. Dissipation at a rate independent of the viscosity, $\varepsilon \sim \nu |\nabla u|^2$ held fixed as $\nu \rightarrow 0$, gives (2); and a structure across which the velocity jumps by $O(1)$ therefore has thickness

$$\delta \sim |\nabla u|^{-1} \sim \nu^{1/2}. \quad (3)$$

At $\nu = 10^{-6}$ that is a thousandth of the macroscopic scale; at 10^{-12} , a millionth. As $\nu \rightarrow 0$ the structure thins without limit while the velocity jump across it stays bounded: the gradient diverges, so the solution is not smooth, and the velocity does not, so it is not singular. That is the whole of the third case in one line, and (3) is the reason no amount of regularity theory reaches it.

It also disposes of a tempting misreading. A singular structure produced by letting $\nu \rightarrow 0$ is not a counterexample to (1) at any fixed viscosity, because by (3) its thickness at that viscosity is $\nu^{1/2}$ and finite, and the solution there is bounded. Such a structure is singular only in the limit, where the equation being solved is Euler's and not Navier–Stokes's. The Clay problem is posed at fixed positive viscosity. A construction whose singularity shrinks to nothing as the viscosity is removed has answered the Euler question — itself open, and interesting — and not the one asked. The same test applies to any claimed blowup: does the singular structure survive at fixed $\nu > 0$ with data satisfying the stated decay, or does it exist only in a limit, or only with a forcing term the formulation excludes?

3 What should replace it: wellposedness

The defect in the official formulation is not that it asks a hard question. It is that it asks about the wrong property. Smoothness is not what makes a solution concept physically meaningful. Wellposedness is.

Hadamard's criterion has three parts: a solution exists, it is unique, and it depends continuously on the data. The third is the one that carries the physical content, because data are never exact. A problem whose solution is not stable under perturbation of the data is not a model of anything measurable, whatever its smoothness.

Weak solutions exist. Leray showed in 1934 that weak solutions of (1) exist globally in time [2]. What was left open was not existence but whether the weak solution concept is *wellposed* — whether it determines anything. That is the question worth asking, and it is not the question the Clay problem asks.

Wellposedness is a property of outputs, not of solutions. Here is the point on which the reformulation turns. For a turbulent solution, the pointwise velocity at a given place and time is not determined: perturb the data in the last decimal and the pointwise field is different everywhere within a short time. In that sense the solution is not wellposed and never will be.

But nobody measures the pointwise velocity of a turbulent flow, and no engineering question depends on it. What is measured, and what is asked for, are *mean-value outputs*: the drag on a body, the lift on a wing, a pressure averaged over a surface, a flux through a section. These are functionals of the solution, and the question of wellposedness must be asked of them:

$$M(u) = \int_Q u \cdot \psi \, dx \, dt \quad (4)$$

for a given weight ψ . The claim is that $M(u)$ is stable for turbulent solutions even though u is not.

The mechanism, and why it inverts the classical expectation. Stability of an output is governed by an associated dual, or adjoint, problem: linearise about the computed solution, run it backward from the output functional, and the growth of the dual solution measures how strongly a perturbation of the data or of the residual can move the output. What the dual problem shows is the reverse of what intuition suggests. Over a *turbulent* velocity field — oscillating rapidly in space and time — the dual solution grows only modestly, because the linearised coefficients oscillate and their effect largely cancels. Over a *smooth* field the dual solution grows, because there is nothing to cancel. Turbulence stabilises the outputs that smoothness destabilises.

This is the sense in which the reformulated problem has an answer, and the answer is not the one the dichotomy anticipates. Smooth solutions are not wellposed. Turbulent solutions are wellposed in mean-value outputs. The physically meaningful solution concept is the one the official formulation treats as the failure case.

4 The reformulated problem

We state the replacement plainly.

Problem. For the incompressible Navier–Stokes equations at small viscosity, with smooth divergence-free initial data, show that the weak solution is *wellposed in mean-value outputs*: that functionals of the form (4) depend continuously on the data, with quantitative error control, even though the pointwise solution does not. Equivalently: establish the second law of thermodynamics for fluids — that a turbulent solution dissipates kinetic energy at a rate which does not vanish with the viscosity, and that this dissipation is what makes the outputs stable.

Two features of this statement are worth marking. It contains the viscosity, so it is a statement about fluids and not about a normalised abstraction. And it asks for stability rather than regularity, so it can be answered affirmatively for the solutions that actually occur instead of being answered vacuously for solutions that do not.

5 In what sense this constitutes a solution

The solutions are constructed, not assumed. A finite element method for (1) with residual-based stabilisation — Galerkin least squares on a mesh, piecewise polynomial velocity and pressure, no turbulence model supplied by the user (§6) — produces solutions which become turbulent even when initiated from smooth potential data, and which stay bounded. The stabilisation supplies dissipation in proportion to the residual, so the dissipation is generated by the computation rather than prescribed by a model. In the vanishing-viscosity limit the same scheme is an Euler solver whose turbulent dissipation appears automatically. The non-smooth, non-singular third case is therefore not a conjecture: it is what the equations produce when they are solved.

The residual behaves as the theory requires. For such solutions the Navier–Stokes residual $R(U, P)$ is small in a weak norm and large in a strong one. Small weakly is what makes the computed field a solution; large strongly is the statement that it is not smooth. A method whose residual were small in both would be computing something that does not exist.

The error is controlled by duality, not by asserted convergence. An a posteriori error estimate of the form

$$|M(u) - M(U)| \leq \int_Q R(U, P) \cdot \varphi \, dx \, dt \leq S \|R(U, P)\|_{-1}, \quad (5)$$

with φ the dual solution and S its stability factor, converts a computed solution into a statement with a bound attached. The stability factor is not assumed; it is computed by solving the dual problem. It is through (5) that the wellposedness claim becomes quantitative: S is moderate for mean-value outputs over turbulent solutions and large for pointwise outputs and for smooth solutions, which is the distinction of §3 expressed as a number.

Computation as proof. That an answer rests on computation is not a defect peculiar to this problem. The four-colour theorem was settled the same way, and the settlement was accepted because the computation was reproducible and its error controlled. The same standard applies here, and (5) is what meets it: the claim is not that a particular simulation looked turbulent, but that the output of the computation comes with a bound which can be evaluated, and which is small.

6 The computational record

The argument of §5 is not offered as a programme. The method, the error control and the computations are set out in *Computational Turbulent Incompressible Flow* [4], and it is worth stating what that book establishes, because the reformulation of §4 asks for exactly the things it supplies.

G2: the method is Galerkin, and the turbulence model is the residual. General Galerkin, or G2, is a stabilised finite element method — continuous piecewise linear in space and time, cG(1)cG(1), with least-squares control of the residual. The stabilisation is not a subgrid model: no eddy viscosity is prescribed, no wall function, no closure, no constant fitted to measurement. The dissipation is whatever the residual makes it, which means it is generated by the flow being computed rather than supplied by the person computing it. This matters for the present argument specifically. A computation carrying a fitted turbulence model could not be used to say what Navier–Stokes contains, because the turbulence would have been put in by hand. G2 has nothing put in by hand, so what comes out is a property of (1).

The dual problem is solved, and the stability factor is a computed number. The estimate (5) is the book’s organising device. For each output of interest the dual problem is linearised about the computed solution, solved backwards, and its stability factor evaluated; the mesh is then refined where the product of residual and dual weight is largest. The consequence is that the error in the output is not estimated by comparing successive meshes and hoping, but bounded by a quantity the computation itself produces. Adaptivity and error control are the same mechanism.

Mean-value outputs converge; the flow field does not. The book’s central computational finding is the one §3 needs. Mean drag and mean lift for turbulent flow past a bluff body come out to a tolerance of a few per cent on meshes of a few hundred thousand points — small enough to run on a personal computer, and orders of magnitude coarser than a direct simulation resolving the Kolmogorov scale would require. The pointwise velocity field on those meshes is not converged

and does not become so. Both facts hold simultaneously, and together they are the wellposedness claim made quantitative: the outputs are stable because the stability factor for them is moderate, and the pointwise field is not because its stability factor is not.

Turbulent Euler solutions and d’Alembert’s paradox. Taking $\nu \rightarrow 0$ in G2 gives a solver for the Euler equations whose dissipation does not vanish with the viscosity, which is the second law of §4 in computational form. What such solutions produce is not potential flow. A bluff body in this limit shows potential flow modified by a three-dimensional rotational separation, and that separation carries drag — which is the resolution of d’Alembert’s paradox offered in [5]: the paradox arises from taking the smooth potential solution to be the vanishing-viscosity limit, and the correct limit is turbulent. The same analysis applied to a wing gives lift and drag without a circulation condition imposed by hand [6].

This is the concrete content behind the claim of §2 that smooth solutions are unstable. It is not an assertion about what would happen to a perturbed smooth solution; it is a record of what happens when the equations are solved from smooth data without a model to keep them smooth.

7 What the answer says about the original question

To the official question — do smooth data give solutions that remain smooth for all time — the answer implied by the above is no, and not by blowup. Smooth data give solutions that become turbulent: non-smooth, with gradients growing as $\nu^{-1/2}$, bounded in velocity, dissipating at a rate which does not vanish as $\nu \rightarrow 0$. Blowup, the only failure mode the official formulation admits, does not occur and is not the interesting question. The interesting question is the one the formulation leaves out.

This does not claim the prize, and it is not intended to. It claims something less procedural and more useful: that the problem as posed cannot be answered in a way that illuminates fluid mechanics, because the alternative it offers is not exhaustive over the cases that occur; and that the adjacent, answerable question — wellposedness of mean-value outputs of turbulent weak solutions — is both the physically meaningful one and one to which the answer is affirmative and computable.

References

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